Advanced Hyperspectral Image Classification via Spectral–Spatial Redundancy Reduction and TokenLearner-Enhanced Transformer

Jinbin Wu, Jiankang Zhao, and Haihui Long

***Abstract*— Currently, hyperspectral image (HSI) classification methods based on deep learning (DL) are extensively researched. However, the unique imaging characteristics of HSIs introduce a significant amount of spectral–spatial information redundancy, which creates challenges for existing methods in fully processing and utilizing this information. Furthermore, the classical patch-based data augmentation introduces heterogeneous pixels, posing a risk of losing central target information. To tackle these challenges, we propose a novel HSI classification network via spectral–spatial redundancy reduction and TokenLearner-enhanced transformer, named SSRRTL. Leveraging the inherent advantages of convolutional neural networks (CNNs) and transformer architectures, we innovatively introduce three key modules: spectral redundancy reduction and multiscale information extraction (SRRMIE) module, gated spatial feature enhancement (GSFE) module, and TokenLearner module. The SRRMIE module focuses on extracting multiscale features, reducing spectral redundancy, and preserving discriminative features of minor components through separation, feature extraction (FE), and adaptive fusion strategies. To further handle spatial information redundancy, the GSFE module employs a gated mechanism to enhance the extraction of key spatial features and suppress the expression of redundant spatial features. In addition, the TokenLearner method within the transformer architecture integrates feature tokens with central target tokens to enhance semantic feature expression and stabilize central target information, thereby improving the model's robustness. Experimental results on multiple datasets confirm the effectiveness of SSRRTL and its superiority over other state-of-the-art methods.**

***Index Terms*— Adaptive fusion, feature redundancy, hyperspectral image (HSI) classification, self-attention, TokenLearner.**

I. Introduction

With the rapid development of remote sensing technology, hyperspectral imaging has emerged as a crucial tool in various domains such as environmental monitoring, food safety, and agricultural assessment [1], [2], [3]. Hyperspectral images (HSIs) contain rich spectral and spatial information, amplifying their potential for accurate classification [4]. However, the low spatial resolution, extremely high spectral dimensions, and interband similarity also introduce a significant amount of feature redundancy, presenting numerous challenges [5]. Effectively addressing redundant features without losing discriminative features necessary for precise classification stands as a fundamental challenge in HSI processing.

Traditional HSI classification methods primarily rely on machine learning algorithms for spectral feature extraction (FE), such as support vector machine (SVM) [6], k-means clustering [7], and random forest [8]. These methods mainly focus on spectral sequence data and tend to neglect the spatial information. Therefore, a series of morphological operations are applied to extract shape and structural characteristics from HSIs [9]. Nevertheless, conventional methods typically require a large amount of prior knowledge and manual feature engineering, which hinders classification accuracy and efficiency.

Recently, the rise of deep learning (DL) technology presents new prospects for HSI classification [10]. Researchers have developed a range of backbone networks, including recurrent neural networks (RNNs) [11], convolutional neural networks (CNNs) [12], generative adversarial networks [13], graph neural networks [14], transformers [15], and the mamba network [16]. Among these, CNNs and transformers are the most widely used. CNNs excel in extracting local spectral–spatial features due to their weight-sharing mechanism [17]. In contrast, transformers leverage the self-attention mechanism to effectively capture global context by modeling long-range dependencies [18]. Beyond single-network architectures, hybrid networks have been proposed to combine the strengths of different frameworks. Notably, the CNN-transformer architecture has gained significant attention by integrating the local FE capabilities of CNNs with the global feature modeling power of transformers. This synergy enhances feature representation and improves overall classification performance.

Due to the intrinsic characteristics, HSIs contain substantial spectral–spatial feature redundancy, which negatively impacts classification performance and increases computational costs. To mitigate spectral redundancy, mainstream networks typically employ dimensionality reduction techniques such as principal component analysis (PCA) during preprocessing to directly reduce the number of bands. However, such methods may remove band information that is useless for mathematical compression but plays a discriminative role in classification, especially in precise classification scenarios with high interclass spectral similarity, potentially leading to inferior results compared to using the original data [19]. The issue of spatial redundancy also remains to be addressed. On the other hand, existing methods often employ data augmentation

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The authors are with the Department of Instrument Science and Engineering, School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: wujinbin@sjtu.edu.cn; zhaojiankang@sjtu.edu.cn; longhh@sjtu.edu.cn).

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to characterize the central pixel through a patch of surrounding pixels. Conventional networks generally treat all pixels in the patch equally, which is not conducive to improving classification performance. Given the spatial coherence, pixels near the target pixel contain more relevant information and should be emphasized, while edge pixels may carry useless or even misleading data and should receive less attention. Moreover, the extraction of multiscale information and the lightweight design of the network also need to be considered.

In response to the challenges outlined above, we introduce an HSI classification network based on the CNN-transformer architecture, named SSRRTL. This network preserves the CNN-transformer architecture's capacity to capture local contextual features and handle global correlations. To address the redundancy of spectral–spatial features, we design two modules to diminish redundant features while retaining certain classification discriminative information. In addition, we introduce a novel TokenLearner approach to improve feature learning and emphasize pixels in the central region. Furthermore, SSRRTL considers the extraction of multiscale information from HSIs and incorporates lightweight design principles in both multiscale information extraction and the TokenLearner design. The main contributions of this article are summarized as follows.

1) We propose a spectral redundancy reduction and multiscale information extraction (SRRMIE) module. This module preserves minor component classification discriminative information and reduces spectral feature redundancy through operations of separation, FE, and adaptive fusion. With the design of lightweight multiscale convolutions, the SRRMIE module also possesses the capability to extract multiscale features.
2) We propose a gated spatial feature enhancement (GSFE) module that enhances the extraction of information-rich spatial features while suppressing the expression of spatially redundant features through a gating mechanism.
3) We propose a TokenLearner approach that combines feature tokenization with central region tokens. This method aims to enrich the representation of profound semantic features and enhance the network's focus on the target pixel, thereby improving the model's robustness.
4) Extensive experiments conducted on multiple real datasets validate the superior classification performance and robustness of the proposed SSRRTL compared to other state-of-the-art HSI classification methods.

The remainder of this article is organized as follows. Section II introduces the related work. Section III details the proposed SSRRTL network. Section IV reports the experimental results and analysis on several datasets. Finally, the overall conclusion is presented in Section V.

II. Related Work

A. CNN for HSI Classification

In early studies, CNNs used for HSI classification mostly employed a single convolutional form. Hu et al. [20] proposed a 1-D CNN with a five-layer structure to capture spectral features. Makantasis et al. [21] combined PCA with a 2-D CNN to capture spatial contextual features. Chen et al. [22] proposed a 3-D CNN that simultaneously handles spectral–spatial information. To overcome the limitations of the single convolutional form, Roy et al. [23] developed a hybrid CNN combining 2-D and 3-D convolutions. Alkhatib et al. [24] proposed a multibranch CNN that integrates three types of convolutions. Classical hierarchical CNNs can expand the receptive field through pooling layers, but pooling operations destroy the original spatial features and cause information loss. Therefore, expansion convolutional networks [25], mixed depthwise convolutional networks [26], and feature fusion networks [27], [28] are proposed to capture multiscale features. Nevertheless, these networks encounter challenges in extracting long-range contextual features and lack global perception capabilities due to the limitations of convolutional operations.

B. Transformer for HSI Classification

Unlike CNNs, the transformer emphasizes the comprehension of global contextual information. He et al. [29] first applied the transformer to HSI classification by directly converting pixels into sequences and inputting them into a bidirectional encoder to extract sequence features. Hong et al. [15] proposed SpectralFormer, which incorporates groupwise spectral embeddings and cross-layer skip connections inspired by the transformer architecture. Mei et al. [30] proposed a group-aware hierarchical transformer network, which strengthens the learning of local features through grouped pixel embedding. Scheibenreif et al. [31] combined the masking mechanism with the transformer to develop a self-supervised network. Wang et al. [32] introduced a transformer-based dual-context network that converts HSIs into three spectral images to utilize pretrained networks for FE and then extracts regional features through a carefully designed dual-context module. Although some methods are developed to improve the local feature perception ability of the transformer, it still lags behind CNNs in this aspect.

C. CNN-Transformer for HSI Classification

To leverage the inherent advantages of CNNs and transformers, the joint network architecture of CNN-transformer gains wide attention. Roy et al. [33] enhanced the interaction between the class token and the shape information of HSIs through morphological convolution operations. Sun et al. [34] used the PCA algorithm for preprocessing, extracting shallow features through hierarchical hybrid convolutions, and introducing a Gaussian-weighted feature tokenizer to improve the transformer's FE capability. Similarly, Zhang et al. [35] introduced convolutional blocks to extract local information and reduce the computation by designing a lightweight attention mechanism. Zhao et al. [36] introduced lightweight convolution computations in the CNN and reduced the computational effort of the transformer through groupwise separable multihead attention. Although various CNN-transformer-based HSI classification networks are proposed, they poorly address spectral–spatial feature redundancy, limiting classification performance. The direct utilization of the PCA algorithm for dimensionality reduction leads to information loss and the

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potential omission of crucial discriminative details. In addition, the class token used for classification in these methods does not sufficiently prioritize central pixels, which may result in the oversight of critical target information.

III. Methodology

A. Overview of the Proposed SSRRTL

In this work, our proposed SSRRTL maintains the excellent performance of the CNN-transformer architecture. SSRRTL aims to reduce spectral–spatial feature redundancy while preserving minor component classification discriminative information. It facilitates the extraction of representative and multiscale features and directs the network's attention toward classification objectives. The overall structure of SSRRTL is presented in Fig. 1, mainly comprising four modules: SRRMIE, GSFE, TokenLearner, and transformer encoders, which sequentially process HSI data. First, the HSI data passes through the SRRMIE module to mitigate the redundancy of spectral information and extract multiscale features by utilizing the separation-FE-adaptive fusion operation. Subsequently, the GSFE module diminishes spatial redundancy and amplifies crucial spatial features. Finally, the TokenLearner-enhanced transformer strengthens the interaction between the class token and target pixel feature to improve classification performance.

B. Spectral Redundancy Reduction and Multiscale Information Extraction

The original HSI data $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$, where H and W represent the spatial dimensions and B represents the number of bands, undergoes preliminary processing with the PCA algorithm before the SRRMIE module. It is particularly important to emphasize that we use the PCA algorithm not for dimensionality reduction. Although such dimensionality reduction techniques can discard bands that are deemed mathematically insignificant for compression purposes, these bands may contain crucial discriminative information for classification tasks. Particularly in classification scenarios characterized by high interclass spectral similarities, direct compression might lead to spectral information that is almost identical for different categories, potentially resulting in the loss of discriminative details present in the discarded bands. Therefore, we use the PCA algorithm for preprocessing solely to perform a rough prior component ranking, and the resulting $\boldsymbol{X}_{\text{pca}} \in \mathbb{R}^{H\times W\times B}$ still retains the original number of bands. To fully utilize the spectral–spatial information, each target pixel is characterized by a patch $\boldsymbol{X}_1 \in \mathbb{R}^{s\times s\times B}$ of window size $s \times s$ centered around it.

After the input of $\boldsymbol{X}_1$ into the SRRMIE module, the channels are initially divided equally into two parts. The top-ranked $L = B/2$ bands are identified as the significant information stream $\boldsymbol{X}_{\text{up}}$, while the remaining bands are considered as the minor information stream $\boldsymbol{X}_{\text{low}}$. Then, $\boldsymbol{X}_{\text{up}}$ and $\boldsymbol{X}_{\text{low}}$ pass through convolutional layers for FE. As shown in Fig. 1, for $\boldsymbol{X}_{\text{up}}$, we design a lightweight multiscale FE convolution block that combines depthwise convolution (DWC) and pointwise convolution (PWC) to extract rich features. The DWC assigns a convolutional kernel to each channel, aiding in the extraction of local features. Convolutional kernels with three distinct receptive fields are employed to capture multiscale features, followed by a concatenation of the outcomes postbatch normalization (BN) and activation. Subsequently, the PWC with a kernel size of 1×1 is used to aggregate channel information and multiscale features, restoring the number of channels to L. Finally, the residual connection is incorporated to enhance the stability of FE, resulting in $\boldsymbol{Y}_{\text{up}} \in \mathbb{R}^{s\times s\times L}$. Compared to conventional convolution, the combination of DWC and PWC significantly reduces the computational load. For $\boldsymbol{X}_{\text{low}}$, which contains relatively less information, we use the most cost-effective PWC and obtain $\boldsymbol{Y}_{\text{low}} \in \mathbb{R}^{s\times s\times L}$ through the residual connection.

Following the FE process, the output features from the upper and lower branches are adaptively fused. First, global average pooling is applied individually to $\boldsymbol{Y}_{\text{up}}$ and $\boldsymbol{Y}_{\text{low}}$ to derive feature vectors $\boldsymbol{b}_{\text{up}}$ and $\boldsymbol{b}_{\text{low}} \in \mathbb{R}^L$, which encapsulate global information. Then, $\boldsymbol{b}_{\text{up}}$ and $\boldsymbol{b}_{\text{low}}$ are concatenated and activated by softmax function to generate $\boldsymbol{c} \in \mathbb{R}^{2L}$, as follows:

$$\boldsymbol{c} = \text{Softmax}\left[\left(\text{Pooling}\left(\boldsymbol{Y}_{\text{up}}\right),\ \text{Pooling}(\boldsymbol{Y}_{\text{low}})\right)\right] \tag{1}$$

where $[\cdot]$ denotes concatenation. After separation, we obtain the representation vectors $\boldsymbol{c}_{\text{up}}$ and $\boldsymbol{c}_{\text{low}} \in \mathbb{R}^L$, which reflect the feature importance of the two branches. $\boldsymbol{c}_{\text{up}}$ and $\boldsymbol{c}_{\text{low}}$ serve as channel weight vectors to assign weights to the channels of $\boldsymbol{Y}_{\text{up}}$ and $\boldsymbol{Y}_{\text{low}}$, and the weighted sum is obtained as the fused feature $\boldsymbol{Y}_1 \in \mathbb{R}^{s\times s\times L}$, expressed as follows:

$$\boldsymbol{Y}_1 = \boldsymbol{c}_{\text{up}}\boldsymbol{Y}_{\text{up}} + \boldsymbol{c}_{\text{low}}\boldsymbol{Y}_{\text{low}}. \tag{2}$$

The $\boldsymbol{Y}_1$ obtained through the SRRMIE module not only encapsulates multiscale information from the primary components of HSI data but also reduces spectral channel redundancy via adaptive feature fusion. In addition, this process ensures that discriminative features from the minor components are retained, thereby minimizing information loss. The FE process in the SRRMIE module also incorporates lightweight convolutional computations to decrease computational costs.

C. Gated Spatial Feature Enhancement

HSI data encompass not only substantial spectral redundancy but also significant spatial redundancy. To mitigate the redundancy of spatial features and improve the identification of crucial spatial features, the GSFE module is devised. First, the classic spatial attention (SA) block [37] shown in Fig. 2 is used to recalibrate spatial information and amplify spatial distinctions among pixels, resulting in $\boldsymbol{Y}_2$, expressed as

$$\boldsymbol{Y}_2 = \boldsymbol{Y}_1 \otimes \phi\left(f^{7\times 7}\left[\text{AvgPool}(\boldsymbol{Y}_1),\ \text{MaxPool}(\boldsymbol{Y}_1)\right]\right) \tag{3}$$

where AvgPool and MaxPool denote average pooling and max pooling, respectively. $f^{7\times 7}$, ϕ, and $\otimes$ represent a 2-D convolutional layer with a kernel size of 7×7, the activation function LeakyReLU, and pointwise multiplication, respectively.

Next, group normalization (GN) [38] is applied to $\boldsymbol{Y}_2$, and the scaling factors in GN are utilized to evaluate the information richness of various spatial feature maps, expressed as

$$\text{GN}(\boldsymbol{Y}_2) = \boldsymbol{\alpha}\frac{\boldsymbol{Y}_2 - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \boldsymbol{\beta} \tag{4}$$

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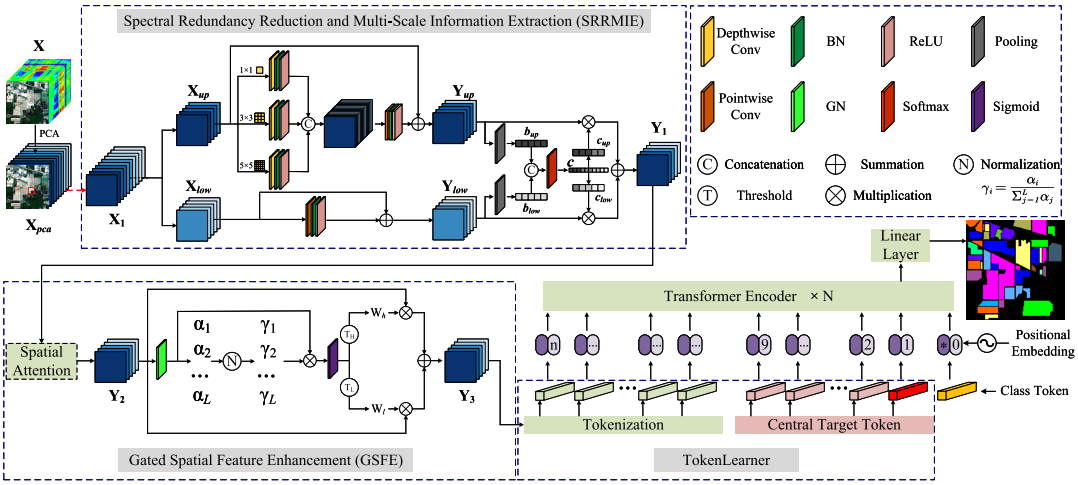


Fig. 1. Proposed SSRRTL network for HSI classification.

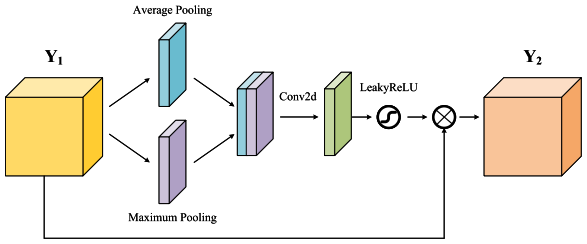


Fig. 2. SA block in GSFE module.

where μ and σ represent the mean and standard deviation, ε is a small constant, and $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are trainable affine transformation parameters. Since richer spatial information reflects more variation in spatial pixels, it results in a larger $\boldsymbol{\alpha}$ [39]. Therefore, we use the trainable parameter $\boldsymbol{\alpha} = \{\alpha_i\} \in \mathbb{R}^L$ to quantify the richness of spatial features and normalize it to obtain $\boldsymbol{\gamma} = \{\gamma_i\} \in \mathbb{R}^L$, which represents the importance of different feature maps, as follows:

$$\gamma_i = \frac{\alpha_i}{\sum_{j=1}^{L} \alpha_j}. \tag{5}$$

We use $\boldsymbol{\gamma}$ to reweight the feature maps and map the results to the range $(0, 1)$ using the sigmoid function. To enhance the expression of rich spatial features and reduce spatial feature redundancy, we design a dual-branch high-low threshold control block. The high and low threshold values are set to 0.75 and 0.25, respectively. For the upper branch, weights exceeding the high threshold are set to 1, obtaining informative $\boldsymbol{W}_h$ to reinforce the expression of rich features. Conversely, weights falling below the low threshold in the lower branch are set to 0, obtaining less informative $\boldsymbol{W}_l$ to decrease feature redundancy. Then, $\boldsymbol{Y}_2$ is reweighted by $\boldsymbol{W}_h$ and $\boldsymbol{W}_l$, respectively, and the results of the two branches are summed and averaged to obtain the final output $\boldsymbol{Y}_3$ of the GSFE module. The complete process of the GSFE module can be represented as follows:

$$\begin{cases} \boldsymbol{Y}_2 = \mathrm{SA}(\boldsymbol{Y}_1) \in \mathbb{R}^{s\times s\times L} \\ \boldsymbol{W} = \mathrm{Gate}(\mathrm{Sigmoid}(\boldsymbol{\gamma}(\mathrm{GN}(\boldsymbol{Y}_2)))) \in \mathbb{R}^{s\times s\times L} \\ \boldsymbol{Y}_3 = \dfrac{1}{2}(\boldsymbol{W}_h \otimes \boldsymbol{Y}_2 + \boldsymbol{W}_l \otimes \boldsymbol{Y}_2) \in \mathbb{R}^{s\times s\times L}. \end{cases} \tag{6}$$

D. TokenLearner

The feature maps obtained after removing spectral–spatial redundancy and extracting local features are passed through the transformer's encoder to better capture global features and deeply mine spectral sequence information. Before this, to further capture high-level semantic features and increase the network's focus on the central pixel to obtain more target classification information and improve the model's robustness, we propose a TokenLearner method that combines feature tokenization with central semantic tokens. First, feature tokenization is applied to convert the feature map $\boldsymbol{Y}_3$ into semantic tokens. $\boldsymbol{Y}_3$ is first flattened to produce s^2 initial tokens and the token dimension is transformed to 64 by the weight matrix $\boldsymbol{W}_a \in \mathbb{R}^{L\times 64}$, denoted as follows:

$$\begin{cases} \boldsymbol{X}_{\mathrm{in}} = \mathrm{Flatten}(\boldsymbol{Y}_3) \in \mathbb{R}^{s^2\times L} \\ \boldsymbol{X}_a = \boldsymbol{X}_{\mathrm{in}} \cdot \boldsymbol{W}_a \in \mathbb{R}^{s^2\times 64}. \end{cases} \tag{7}$$

Another weight matrix $\boldsymbol{W}_b \in \mathbb{R}^{L\times n}$ is used to transform the feature map into a high-level semantic space, activated by the softmax function to focus on important semantics, resulting in $\boldsymbol{X}_{\mathrm{fe}} \in \mathbb{R}^{n\times 64}$ containing n semantic tokens, as follows:

$$\boldsymbol{X}_{\mathrm{fe}} = \mathrm{Softmax}(\boldsymbol{X}_{\mathrm{in}} \cdot \boldsymbol{W}_b)^{\mathrm{T}} \cdot \boldsymbol{X}_a. \tag{8}$$

The essence of feature tokenization is to learn key features based on the integration of all token features through two learnable weights $\boldsymbol{W}_a$ and $\boldsymbol{W}_b$, resulting in n semantic tokens. These two weights are initialized with a standard normal distribution to guide the feature distribution more regularly [34]. Notably, these n semantic tokens are extracted from the overall information of the input feature $\boldsymbol{Y}_3$. However, the essence of $\boldsymbol{Y}_3$ is to characterize the central pixel, whose category is determined by the central pixel. Given the lower spatial resolution and larger pixel span in HSIs, a patch may contain multiple categories of ground objects. Relying solely on feature tokenization may lead to the potential loss of information regarding the ground object represented by the central pixel. Due to spatial continuity, pixels adjacent to the target pixel usually carry more target features. To increase the network's attention on the target pixel and enhance robustness, we restore $\boldsymbol{X}_a$ to its original spatial distribution and

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retain the central tokens $\boldsymbol{X}_{\text{ct}} \in \mathbb{R}^{9\times 64}$ that signify the central 3×3 area.

In addition, a randomly initialized, trainable class token $\boldsymbol{X}_{\text{cls}}$ is used to aggregate global features and make the final classification decision. Finally, trainable positional embeddings $\boldsymbol{X}_{\text{pos}}$ are incorporated into all tokens and merged to yield $\hat{\boldsymbol{X}}$, which contains $n + 10$ tokens, as follows:

$$\hat{\boldsymbol{X}} = [\boldsymbol{X}_{\text{cls}}, \boldsymbol{X}_{\text{fe}}, \boldsymbol{X}_{\text{ct}}] + \boldsymbol{X}_{\text{pos}}. \tag{9}$$

Typically, $n + 9 < s^2$. This TokenLearner approach not only enhances semantic expression and improves the interaction between the class token and central target features but also effectively reduces the computational load of the transformer encoder by reducing the number of tokens.

E. Transformer Encoder

The transformer encoder is used to model the deep relationships between semantic tokens. This module primarily comprises a multihead self-attention (MSA) block, a feed-forward neural network (FNN), and layer normalization. During the attention calculation process, the input $\hat{\boldsymbol{X}}$ is linearly transformed into a query matrix $\boldsymbol{Q}$, a key matrix $\boldsymbol{K}$, and a value matrix $\boldsymbol{V}$. The attention weights are obtained by computing the dot product of $\boldsymbol{Q}$ and $\boldsymbol{K}$, followed by scaling and a softmax operation, as expressed by:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\text{T}}}{\sqrt{d_k}}\right) \tag{10}$$

where d_k is the dimension of $\boldsymbol{K}$. To capture information from different subspaces, $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are usually split into multiple heads, with each head calculating attention scores in parallel. The outputs of these heads are concatenated and linearly transformed to obtain the MSA score, expressed as

$$\text{MSA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = [\text{head}_1, \text{head}_2, \ldots, \text{head}_h]\boldsymbol{W} \tag{11}$$

where h is the number of heads and $\boldsymbol{W}$ is the weight matrix of the linear transformation. The output of the ith head can be calculated by the following formula:

$$\text{head}_i = \text{Attention}\left(\boldsymbol{Q}\boldsymbol{W}_i^{Q}, \boldsymbol{K}\boldsymbol{W}_i^{K}, \boldsymbol{V}\boldsymbol{W}_i^{V}\right) \tag{12}$$

where $\boldsymbol{W}_i^{Q}$, $\boldsymbol{W}_i^{K}$, and $\boldsymbol{W}_i^{V}$ represent the weight matrices of the query, key, and value of the ith head, respectively.

Next, the features are processed through the FFN, which consists of two fully connected layers and a nonlinear activation function, to further enhance the model's perception and expressive capabilities. The result after FFN processing is combined with the output of the MSA block, further enriching the feature representation. Finally, the class token, which integrates global features, passes through a linear layer and uses the softmax function to calculate the probability that the input belongs to a certain category. The sample's classification result is determined based on the highest probability value.

IV. Experiments and Analysis

This section introduces the well-known HSI datasets and experimental conditions used in the experiments. Meanwhile, extensive experiments and ablation analyses are conducted to evaluate the performance of the proposed SSRRTL and compare it with other state-of-the-art methods.

A. Data Description

1) Indian Pines: The Indian Pines (IP) dataset comprises 145×145 pixels with a ground sampling distance (GSD) of 20 m, covering 224 spectral bands ranging from 400 to 2500 nm. Typically, 200 effective bands are used after removing bands affected by noise and water vapor. This dataset consists of 10 249 labeled pixels, distributed across 16 classes.

2) Houston2013: The Houston2013 (HU) dataset covers the University of Houston campus and its neighboring rural areas in the USA. This dataset contains 349×1905 pixels with a GSD of 2.5 m, of which 15 029 labeled pixels are divided into 15 categories. HU records 144 spectral bands within the spectral range of 364–1046 nm, with a spectral resolution of 10 nm.

3) Pavia University: The Pavia University (PU) dataset captures the HSI of the University of Pavia and its surrounding areas in Italy. The PU image contains 610×340 pixels with a GSD of 1.3 m, covering 103 spectral bands within the spectral range of 430–860 nm. This dataset includes a total of 42 776 labeled samples, categorized into 9 classes.

4) Salinas: The Salinas (SA) dataset includes the SA Valley region in California, USA. It comprises 512×217 pixels with a GSD of 3.7 m. The original images capture 224 spectral bands ranging from 400 to 2500 nm, with 204 effective bands retained after filtering. There are a total of 54 129 labeled samples, categorized into 16 classes.

B. Experimental Setup

1) Comparison Methods: We select ten state-of-the-art HSI classification methods for comparative evaluation, including the traditional machine learning algorithm SVM (code from [40]), the classic RNN [41], the hybrid convolution-based backbone CNN (HybridSN [23]), the multibranch hybrid CNN (TriCNN [24]), the CNN-RNN fusion network (CRNN [5]), the transformer-based backbone network (SpectralFormer [15]), and the CNN-transformer fusion networks (MRViT [42], SSFTT [34], morphFormer [33], and GSCViT [36]).

2) Evaluation Metrics: To comprehensively assess the classification performance of different models, three assessment metrics are utilized for quantitative comparison: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). Furthermore, we visualize the classification results to enable a more intuitive qualitative analysis.

3) Implementation Details: For the four datasets IP, HU, PU, and SA, 5%, 5%, 1%, and 0.5% of samples per class are selected for training and ensure that there are no less than 15 training samples for each class. The remaining samples are allocated for testing. Detailed counts of training and testing samples are presented in the experimental results for each dataset. All experiments are conducted on an NVIDIA

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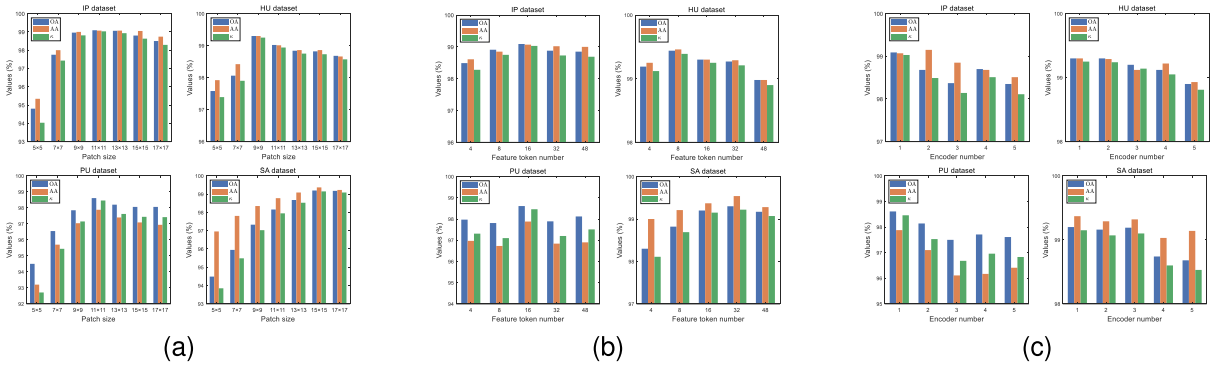


Fig. 3. Impact of different parameters on the performance of SSRRTL. (a) Patch size. (b) Number of feature tokens. (c) Number of encoders.

GeForce RTX 3090 GPU with 24-GB RAM and an i9-10980XE 3-GHz CPU. Our proposed SSRRTL uses the Adam optimizer to update the parameters with the initial learning rate of 1×10^{-3}, the batch size of 64, and training epochs of 100.

C. Parameter Analysis

1) Patch Size: Larger patch sizes contain more contextual pixels, enabling the network to retrieve increased spatial information. However, larger patch sizes are not always advantageous. On one hand, due to the low spatial resolution of HSI data, larger windows may include information from multiple other categories, which can impede the extraction of target features when assigning labels to the central target pixel. On the other hand, larger patch sizes also increase the computational load of the network. Fig. 3(a) illustrates the impact of different patch sizes on model performance. Generally, performance metrics show improvement and stabilization as patch size increases. However, excessively large patch sizes result in a slight decline in classification performance. In addition, the optimal patch size for achieving the best classification performance on the SA dataset is significantly larger than that for the other datasets. This discrepancy is primarily attributed to the SA dataset's extensive smooth regions and densely distributed samples. Consequently, the optimized patch sizes for the four datasets are 11, 9, 11, and 15, respectively.

2) Number of Feature Tokens: Fig. 3(b) shows the impact of the number of feature tokens n used in the TokenLearner module. The optimal n for achieving the best classification performance with SSRRTL varies among different datasets, indicating that the diversity of semantic information in different datasets and even in different regions of the same dataset is challenging to fully express with a fixed number of feature tokens. Moreover, a moderate n consistently provides balanced and excellent performance across all four datasets. When n is too small, the representation capability is limited, making it difficult to capture the complex semantic information. Conversely, when n is too large, irrelevant information may be introduced, affecting classification judgment and increasing computational overhead. Considering the performance across all four datasets, the number of feature tokens is set to 16.

3) Number of Encoders: The impact of the number of transformer encoders N is depicted in Fig. 3(c). Increasing N generally deepens the model, allowing for the capture of more intricate patterns and features. However, this improvement is accompanied by higher memory consumption and computational needs. Experimental findings demonstrate that SSRRTL achieves optimal classification performance across all four datasets when N is approximately 1. Given the current scarcity of training samples for HSI classification, escalating N heightens the risk of overfitting, potentially diminishing decision performance and generalization capability. Since HSIs already contain rich spectral information that can be effectively leveraged by a shallow network, setting N to 1 helps mitigate the dangers of overfitting while minimizing unnecessary memory and computational requirements.

D. Results and Analysis

1) Classification Results and Analysis on the IP Dataset: Table I presents the quantitative results on the IP dataset, with the corresponding classification maps illustrated in Fig. 4. The results indicate that DL-based methods generally outperform the traditional SVM algorithm, which is closely related to the powerful feature learning capabilities of DL. Due to the inability of SVM and RNN to effectively utilize spatial information in HSI data, a large number of misclassifications appear in the classification maps, resulting in lower classification performance. In contrast, the hybrid architectures based on CNN-transformer exhibit superior performance. Notably, the proposed SSRRTL achieves the best results across all three metrics and performs the best in 12 out of the 16 categories.

For this dataset, different categories have relatively similar spectral curves. Methods like MRViT, SSFTT, and morphFormer, which directly use PCA for dimensionality reduction, may compress similar categories of data into the same result, thereby losing potential classification information. Conversely, SSRRTL retains all bands and employs an adaptive fusion strategy in the SRRMIE module to preserve discriminative features from minor components, resulting in stronger information extraction and more comprehensive results.

In addition, the marked area in Fig. 4 contains multiple land cover types, making it one of the most challenging classification regions. Within this confined area, using patch data to enhance the representation of the center pixel may include diverse categories of pixels, potentially introducing misleading information. Therefore, emphasizing the central

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE IP DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

NO.	Class	Train	Test	SVM	RNN	HybridSN	TriCNN	CRNN	SpectralFormer	MRViT	SSFTT	morphFormer	GSCViT	Proposed
1	Alfalfa	15	31	80.65	83.87	**100.00**	87.10	**100.00**	74.19	**100.00**	**100.00**	96.77	**100.00**	**100.00**
2	Corn-notill	71	1357	65.95	67.06	91.45	74.21	82.76	78.11	93.52	94.18	97.27	89.24	**98.75**
3	Corn-mintill	41	789	59.06	66.29	79.21	94.93	85.80	71.23	**99.24**	91.00	96.70	97.97	97.85
4	Corn	15	222	51.80	53.60	87.84	41.44	72.97	75.23	86.94	94.59	98.65	95.50	**100.00**
5	Grass-pasture	24	459	76.91	75.82	92.59	87.58	78.43	78.87	95.64	89.76	97.82	96.95	**99.13**
6	Grass-trees	36	694	94.67	88.76	98.99	**100.00**	93.23	92.22	97.69	98.41	97.12	99.57	99.71
7	Grass-pasture-mowed	15	13	92.31	84.62	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
8	Hay-windrowed	23	455	93.85	87.03	**100.00**	**100.00**	**100.00**	96.48	99.56	**100.00**	**100.00**	**100.00**	**100.00**
9	Oats	15	5	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	Soybean-notill	48	924	66.02	80.19	93.72	75.43	78.46	78.25	89.39	89.07	92.32	89.83	**97.19**
11	Soybean-mintill	122	2333	76.68	76.85	96.49	92.71	96.14	85.08	98.93	98.54	98.93	98.59	**99.83**
12	Soybean-clean	29	564	58.69	50.53	78.55	51.06	82.62	62.59	93.26	97.70	94.86	88.83	**99.11**
13	Wheat	15	190	90.53	94.21	92.11	99.47	94.74	98.95	**100.00**	**100.00**	98.95	98.95	97.89
14	Woods	63	1202	95.34	93.51	97.67	99.25	99.25	91.26	99.42	99.33	99.83	98.17	**99.92**
15	Buildings-Grass-Trees-Drives	19	367	42.23	45.78	79.02	42.23	91.01	46.05	96.19	95.10	83.92	90.46	**98.37**
16	Stone-Steel-Towers	15	78	93.59	83.33	97.44	**100.00**	94.87	**100.00**	93.59	93.59	97.44	98.72	97.44
Overall Accuracy ↑ (OA %)				74.68	75.48	92.45	84.76	89.72	81.21	96.41	95.88	97.04	95.47	**99.09**
Average Accuracy ↑ (AA %)				77.39	76.97	92.82	84.09	90.64	83.03	96.46	96.33	96.91	96.42	**99.07**
Kappa Coefficient ↑ (κ %)				73.12	73.96	91.99	83.83	97.00	80.05	96.18	95.62	96.85	95.19	**99.03**

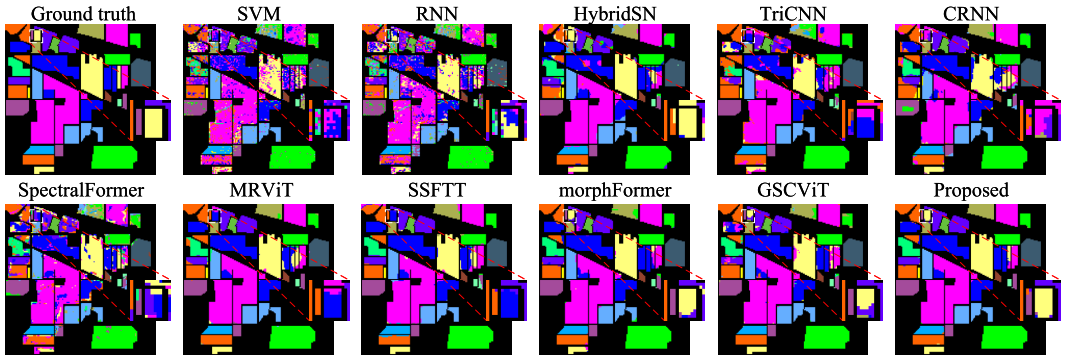


Fig. 4. Classification maps of different methods on the IP dataset.

target information holds paramount importance. As shown in Fig. 4, our method achieves leading performance in this area.

2) Classification Results and Analysis on the HU Dataset: Table II details the experimental results on the HU dataset. Although the HU dataset has a relatively balanced number of labeled samples across different categories, the sample distribution is quite scattered with weak spatial continuity. Nonetheless, the spectral–spatial-based methods still perform better compared to the spectral-based methods. Our proposed SSRRTL achieves the highest scores in OA, AA, and κ, and demonstrates superior performance in 11 out of 15 categories. Even though the spatial information in this dataset is relatively limited, the design of the GSFE module enhances the extraction of effective spatial features and suppresses irrelevant features, thereby improving classification stability. Due to the dispersed nature of the sample distribution, the visual differences in Fig. 5 are not significant. The highlighted area represents a region with a relatively concentrated sample distribution. In this region, most methods achieve satisfactory results, although some methods exhibit minor misclassifications in the boundary areas. The proposed SSRRTL achieves results almost identical to the ground truth in this area.

3) Classification Results and Analysis on the PU Dataset: Table III details the classification results on the PU dataset. All methods based on CNN-transformer architectures achieve an OA above 95%, with the proposed SSRRTL achieving the highest at 98.61%, and also obtaining the best results in AA and κ. On this dataset, different methods exhibit various strengths and weaknesses across different categories. SSRRTL achieves the best results in 4 out of 9 categories and maintains a leading level in the remaining categories. In Fig. 6, two areas are highlighted, with the central red circle area mainly composed of bare soil. These methods experience varying degrees of misclassification, with bare soil being misidentified as meadows, presumably due to their high spectral similarity. However, SSRRTL almost perfectly distinguishes these two highly similar features in this area. Another highlighted area in the lower left corner contains multiple categories, with rapid changes and irregular boundaries, making classification challenging. Nonetheless, SSRRTL still performs excellently owing to its outstanding ability to extract discriminative information, even in complex environments.

4) Classification Results and Analysis on the SA Dataset: Table IV and Fig. 7 present the experimental results. Quantitative results indicate that CNN-based methods outperform those relying solely on RNN and the pure transformer network SpectralFormer, which is closely related to the sample characteristics. The ground truth map in Fig. 7

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TABLE II

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE HU DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

NO.	Class	Train	Test	SVM	RNN	HybridSN	TriCNN	CRNN	SpectralFormer	MRViT	SSFTT	morphFormer	GSCViT	Proposed
1	Healthy grass	62	1189	98.40	92.43	96.47	99.41	96.80	98.15	97.06	99.41	96.64	98.40	**99.75**
2	Stressed grass	62	1192	96.56	97.40	94.55	99.33	92.70	95.72	99.92	**100.00**	99.50	99.08	99.08
3	Synthetic grass	34	663	99.85	98.34	99.70	97.89	93.51	94.42	**100.00**	99.40	99.10	97.29	**100.00**
4	Trees	62	1182	95.69	98.05	97.97	**100.00**	97.88	99.41	98.65	99.49	99.58	98.65	99.24
5	Soil	62	1180	99.07	95.76	**100.00**	99.83	98.56	99.75	**100.00**	**100.00**	98.98	**100.00**	**100.00**
6	Water	16	309	90.61	93.85	85.11	94.82	80.26	89.97	98.06	98.06	99.68	99.03	**100.00**
7	Residential	63	1205	90.87	91.20	89.79	91.37	98.01	94.94	96.76	98.17	97.26	92.20	**99.42**
8	Commercial	62	1182	63.96	77.75	96.02	92.81	85.36	89.76	92.22	93.82	96.02	95.52	**99.24**
9	Road	62	1190	80.34	82.27	87.39	94.79	87.39	88.99	92.35	97.56	90.08	97.06	**99.24**
10	Highway	61	1166	79.67	85.42	98.37	98.71	97.00	94.08	96.91	**100.00**	97.08	98.54	99.40
11	Railway	61	1174	89.61	85.35	88.5	98.64	91.74	93.61	92.16	90.97	**99.15**	97.79	98.47
12	Parking Lot 1	61	1172	52.56	75.77	98.29	91.72	91.04	95.14	89.93	87.88	96.08	93.86	**98.98**
13	Parking Lot 2	23	446	38.79	31.17	88.57	94.62	90.13	62.78	86.32	95.50	85.87	93.72	**96.64**
14	Tennis Court	21	407	98.28	93.12	**100.00**	**100.00**	89.19	95.58	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
15	Running Track	33	627	98.41	98.25	99.68	**100.00**	**100.00**	99.20	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Overall Accuracy ↑ (OA %)				85.11	87.57	94.92	96.85	93.40	94.02	95.89	97.09	97.05	97.26	**99.30**
Average Accuracy ↑ (AA %)				84.84	86.41	94.69	96.93	92.64	92.77	96.02	97.35	97.00	97.41	**99.30**
Kappa Coefficient ↑ (κ %)				84.06	86.69	94.56	96.63	92.93	93.60	95.60	96.89	96.84	97.06	**99.25**

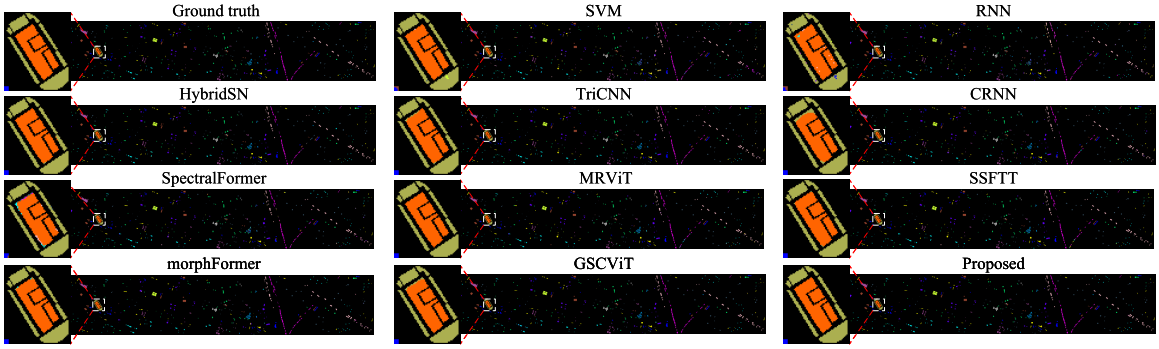


Fig. 5. Classification maps of different methods on the HU dataset.

TABLE III

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE PU DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

NO.	Class	Train	Test	SVM	RNN	HybridSN	TriCNN	CRNN	SpectralFormer	MRViT	SSFTT	morphFormer	GSCViT	Proposed
1	Asphalt	66	6565	89.00	82.96	96.41	91.47	94.38	82.21	96.09	**98.20**	92.41	92.11	97.08
2	Meadows	186	18463	97.68	91.60	99.98	99.57	99.29	98.07	99.98	**100.00**	99.56	99.52	**100.00**
3	Gravel	20	2079	47.81	61.18	83.07	81.87	83.98	72.78	83.69	94.66	83.26	65.42	**96.58**
4	Trees	30	3034	83.03	80.19	89.91	**95.58**	84.54	90.11	88.53	88.89	93.94	91.23	94.59
5	Painted metal sheets	15	1330	98.80	97.29	99.70	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.92	**100.00**
6	Bare Soil	50	4979	69.37	49.21	96.53	83.75	72.04	61.16	98.39	98.29	99.26	94.56	**99.86**
7	Bitumen	15	1315	65.32	84.26	**98.78**	98.63	77.57	65.86	96.35	97.49	90.87	98.63	98.10
8	Self-Blocking Bricks	36	3646	81.27	85.57	87.14	79.37	90.02	74.63	84.50	88.89	95.17	**97.67**	97.07
9	Shadows	15	932	**100.00**	96.89	80.36	98.71	83.58	97.75	96.46	95.71	97.53	98.82	97.64
Overall Accuracy ↑ (OA %)				87.18	82.51	95.89	93.53	91.72	86.49	96.05	97.33	96.53	95.33	**98.61**
Average Accuracy ↑ (AA %)				81.36	81.02	92.43	92.11	87.27	82.51	93.78	95.79	94.67	93.10	**97.88**
Kappa Coefficient ↑ (κ %)				85.90	80.76	95.46	92.87	90.80	85.14	95.64	97.06	96.17	94.85	**98.46**

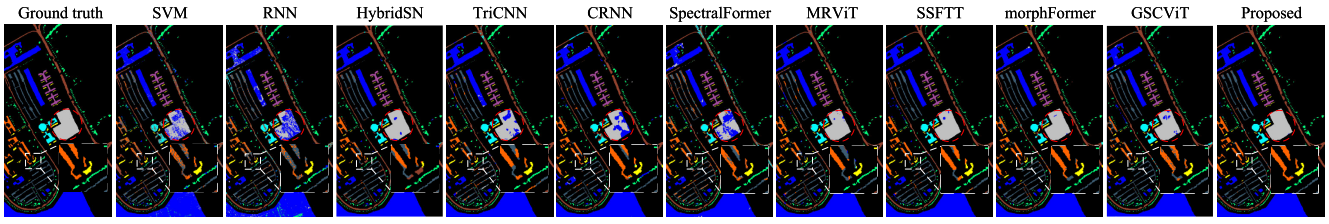


Fig. 6. Classification maps of different methods on the PU dataset.

reveals that the different categories in this dataset have regular boundaries and continuous spatial distribution, providing rich spatial correlation information. Compared to RNNs and transformers, CNNs are adept at handling spatial information, which is beneficial for FE on this dataset. The CNN module in SSRRTL exhibits excellent multiscale spatial

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TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE SA DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

NO.	Class	Train	Test	SVM	RNN	HybridSN	TriCNN	CRNN	SpectralFormer	MRViT	SSFTT	morphFormer	GSCViT	Proposed
1	Brocoli-green-weeds-1	15	1994	97.19	97.39	**100.00**	**100.00**	87.66	98.80	99.50	**100.00**	**100.00**	**100.00**	99.80
2	Brocoli-green-weeds-2	18	3708	94.36	97.63	**100.00**	99.70	99.80	98.71	99.87	**100.00**	**100.00**	**100.00**	**100.00**
3	Fallow	15	1961	98.62	96.84	99.90	96.79	98.88	92.25	**100.00**	**100.00**	87.97	99.24	**100.00**
4	Fallow-rough-plow	15	1379	98.55	93.33	87.60	99.13	99.93	97.90	97.90	**100.00**	98.55	99.93	99.85
5	Fallow-smooth	15	2663	92.53	97.56	**100.00**	95.79	97.86	88.10	98.39	96.73	95.31	95.53	95.79
6	Stubble	19	3940	98.65	98.83	97.01	**100.00**	99.72	99.97	99.26	**100.00**	99.11	99.26	**100.00**
7	Celery	17	3562	99.47	99.07	97.64	99.92	94.86	98.12	**100.00**	98.93	**100.00**	**100.00**	99.78
8	Grapes-untrained	56	11215	90.97	73.40	97.33	91.30	85.53	83.83	91.42	89.53	92.85	82.85	**98.82**
9	Soil-vinyard-develop	31	6172	98.53	97.08	**100.00**	99.72	97.55	97.31	99.68	99.79	99.34	99.51	**100.00**
10	Corn-senesced-green-weeds	16	3262	78.26	86.39	96.11	94.70	85.44	85.81	97.73	96.08	76.82	96.14	**98.96**
11	Lettuce-romaine-4wk	15	1053	91.17	96.87	**100.00**	**100.00**	95.16	95.25	**100.00**	99.24	96.77	99.43	**100.00**
12	Lettuce-romaine-5wk	15	1912	99.69	99.74	99.69	90.64	**100.00**	**100.00**	**100.00**	**100.00**	96.13	98.59	99.84
13	Lettuce-romaine-6wk	15	901	98.56	97.11	99.56	**100.00**	94.78	99.22	99.89	99.45	85.46	96.78	**100.00**
14	Lettuce-romaine-7wk	15	1055	91.75	95.55	99.81	97.35	99.91	99.43	96.78	99.81	96.68	**100.00**	**100.00**
15	Vinyard-untrained	36	7232	42.71	56.89	89.62	77.16	86.70	67.59	94.58	95.08	70.63	86.53	**98.47**
16	Vinyard-vertical-trellis	15	1792	94.81	95.54	97.71	95.48	92.86	89.01	92.75	95.26	96.21	98.27	**98.60**
	Overall Accuracy ↑ (OA %)			87.30	86.33	97.01	93.85	92.64	89.44	96.79	96.48	91.67	93.85	**99.20**
	Average Accuracy ↑ (AA %)			91.61	92.45	97.62	96.10	94.80	93.21	97.98	98.12	93.24	97.00	**99.37**
	Kappa Coefficient ↑ (κ %)			86.51	85.46	96.67	93.46	92.17	88.76	96.58	96.25	91.15	93.45	**99.15**

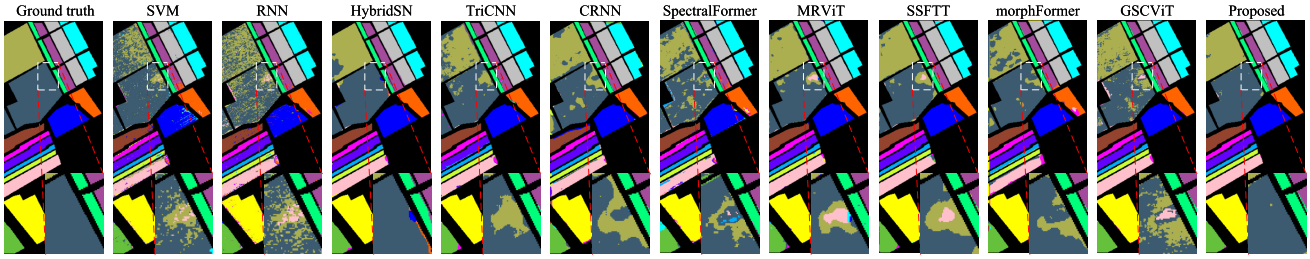


Fig. 7. Classification maps of different methods on the SA dataset.

FE and enhancement capabilities while effectively removing spectral–spatial redundant information, thereby improving classification performance. In addition, SSRRTL combines the advantages of CNN and transformer, integrating spatial information and spectral sequence information, and unifying local and global modeling capabilities, which further enhances classification performance. Consequently, SSRRTL achieves the best OA, AA, and κ, and attains the best results in 11 out of 16 categories. In Fig. 7, the misclassified noise points are mainly concentrated between the two highly similar crops in the upper left corner, with the highlighted area being the most prominent. For classification scenarios with high interclass spectral similarity, the SRRMIE module in SSRRTL integrates the potential discriminative features of all spectral bands, minimizing the loss of discriminative information. This strategy optimally distinguishes between the two crops and demonstrates superior performance in the overall classification map.

5) Classification Results and Analysis on the DFC2018 Dataset: To further validate the performance of the algorithm, we conducted tests on the 2018 IEEE GRSS data fusion contest (DFC) dataset. Unlike the previous datasets, the DFC dataset is large-scale, containing 601 × 2384 pixels and 504 856 labeled points across 20 categories. However, it has fewer spectral bands, only 48. Table V presents the training and testing data for different categories and the quantitative results, while Fig. 8 shows the corresponding visual results. Due to the large data size and diverse categories, the classification task becomes more complex and challenging. The experimental results demonstrate that different methods exhibit their respective strengths across various categories, with our proposed method achieving the best classification performance in 11 out of 20 categories. Furthermore, our method outperforms all comparative methods in terms of the OA, AA, and κ.

6) Impact of Training Sample Ratio: To investigate the robustness of the classification performance of different methods under various training sample ratios, we set up five groups of experiments within the range of 1%–9% sample ratios. To prevent the occurrence of random effects due to an insufficient number of samples in certain categories, we ensure a minimum of 5 and 10 samples per category in the 1% and 3% experiments, respectively. These experiments are mainly conducted on the IP, HU, and DFC datasets. The results depicted in Fig. 9 indicate that with the increase in the training sample ratio, the classification accuracy improves and achieves excellent classification results at higher training sample ratios. Notably, our proposed SSRRTL achieves the best classification results, particularly showing a significant advantage at lower training sample ratios, demonstrating high robustness.

E. Ablation Study

To study the impact of the three modules on the model's performance, we conduct ablation experiments. Table VI presents

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TABLE V

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE DFC DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

NO.	Class	Train	Test	SVM	RNN	HybridSN	TriCNN	CRNN	SpectralFormer	MRViT	SSFTT	morphFormer	GSCViT	Proposed
1	Healthy Grass	489	9310	**97.54**	90.92	90.76	91.06	86.61	68.28	74.24	82.79	96.82	83.50	92.06
2	Stressed Grass	1625	30877	94.68	80.84	92.64	88.23	91.02	**98.25**	95.69	92.58	86.96	97.42	96.72
3	Artificial Turf	34	650	**100.00**	98.46	98.31	**100.00**	98.62	98.00	99.23	96.92	94.62	65.69	97.54
4	Evergreen Trees	679	12916	95.77	94.07	98.21	98.00	95.62	96.03	98.06	96.04	**98.54**	96.38	97.80
5	Deciduous Trees	251	4770	61.84	60.40	88.45	87.51	82.45	92.05	**94.70**	87.95	92.45	80.19	93.77
6	Bare Earth	225	4291	86.20	85.57	99.49	99.09	97.55	99.42	99.42	98.53	98.76	99.63	**100.00**
7	Water	15	251	92.03	95.62	95.62	**100.00**	84.46	66.53	**100.00**	**100.00**	80.08	**100.00**	**100.00**
8	Residential Buildings	1988	37784	78.67	79.33	92.36	90.75	90.73	93.60	97.83	92.86	94.07	85.90	**98.29**
9	Non-residential Buildings	11187	212565	93.55	93.61	98.35	98.89	98.61	98.46	94.58	99.10	**99.58**	96.95	98.66
10	Roads	2293	43573	48.10	62.83	87.19	75.42	75.15	82.79	**94.61**	82.06	87.60	65.91	88.13
11	Sidewalks	1701	32328	48.08	53.67	72.94	73.05	71.21	76.05	76.85	78.82	78.37	66.31	**83.39**
12	Crosswalks	75	1443	0	9.49	25.92	2.70	3.81	12.27	15.8	38.25	8.32	26.33	**38.81**
13	Major Thoroughfares	2317	44031	61.54	60.58	87.86	87.37	75.72	84.77	92.84	92.52	86.25	67.36	**95.62**
14	Highways	493	9372	64.05	71.86	96.64	92.05	95.10	90.91	97.97	84.85	91.67	94.50	**99.24**
15	Railways	346	6591	90.96	90.61	**99.91**	99.53	95.52	99.32	98.00	98.26	99.68	99.82	99.79
16	Paved Parking Lots	575	10925	77.38	74.65	95.10	96.48	58.68	95.64	96.19	97.76	98.67	98.23	**98.76**
17	Unpaved Parking Lots	15	131	75.57	64.12	97.71	95.42	37.40	63.36	98.47	93.89	59.54	88.55	**100.00**
18	Cars	327	6220	23.70	42.73	90.00	86.83	92.44	75.31	96.41	89.90	93.86	96.50	**96.83**
19	Trains	268	5101	59.42	67.42	96.94	99.20	99.12	98.90	99.22	93.18	94.06	96.37	**99.86**
20	Stadium Seats	341	6483	82.54	89.53	99.46	98.57	98.32	99.55	96.59	99.77	98.18	**98.84**	99.80
Overall Accuracy ↑ (OA %)				79.40	80.55	93.17	91.72	89.53	92.40	93.32	93.47	93.81	87.87	**95.90**
Average Accuracy ↑ (AA %)				71.58	73.32	90.19	88.01	81.41	84.48	90.84	89.80	86.90	85.22	**93.75**
Kappa Coefficient ↑ (κ %)				78.55	79.74	92.88	91.37	89.10	92.08	93.04	93.20	93.55	87.36	**95.73**

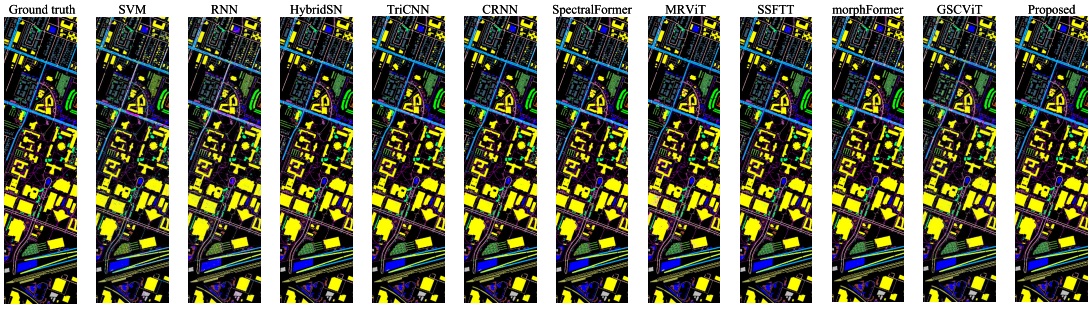


Fig. 8. Classification maps of different methods on the DFC dataset.

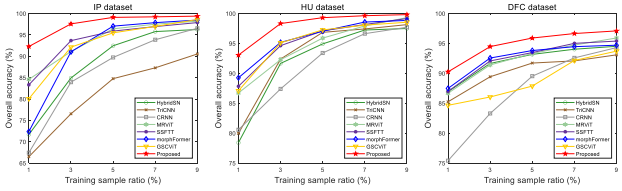


Fig. 9. Classification results under different training sample ratios.

TABLE VI

ABLATION STUDY ON FIVE DATASETS

Module			Overall Accuracy (%)				
SRRMIE	GSFE	TokenLearner	IP	HU	PU	SA	DFC
✗	✗	✗	94.99	96.52	94.77	94.87	88.18
✓	✗	✗	97.50	98.40	96.94	97.45	92.14
✓	✓	✗	98.73	98.85	97.46	98.64	94.03
✓	✗	✓	98.81	98.98	98.29	98.62	94.44
✓	✓	✓	**99.09**	**99.30**	**98.61**	**99.20**	**95.90**

TABLE VII

ANALYSIS OF THE IMPACT OF CENTRAL TARGET TOKENS $\boldsymbol{X}_{\text{ct}}$ ON OA (%) WITH DIFFERENT PATCH SIZES

Dataset	$\boldsymbol{X}_{ct}$	Patch Size			
		5×5	11×11	17×17	23×23
IP	✗	**95.04**	98.18	97.20	96.14
	✓	94.80	**99.09**	**98.50**	**97.68**
HU	✗	97.16	98.21	97.26	96.14
	✓	**97.58**	**99.02**	**98.68**	**98.31**
PU	✗	**95.61**	97.51	96.96	95.26
	✓	94.50	**98.61**	**98.06**	**98.13**
SA	✗	94.08	97.72	98.93	99.02
	✓	**94.49**	**98.33**	**99.18**	**99.39**
DFC	✗	**91.58**	94.19	95.06	94.46
	✓	91.32	**94.55**	**95.56**	**95.88**

TABLE VIII

IMPACT ANALYSIS OF PCA PREPROCESSING

PCA Prior		IP	HU	PU	SA	DFC
OA (%)	✗	96.79	97.56	97.09	98.22	95.78
OA (%)	✓	**99.09**	**99.30**	**98.61**	**99.20**	**95.90**

the experimental results with different module combinations. The results indicate that the inclusion of the three modules improved the model's performance by approximately 7.5% on the more challenging DFC dataset, and by about 4% on other smaller-scale datasets. Specifically, the SRRMIE module processes the complete spectral information of HSIs. By utilizing multiscale FE and adaptive feature fusion, this module effectively reduces spectral information redundancy while maximizing the preservation of discriminative features necessary for classification. Therefore, this module contributed the most to the performance improvement, with a gain of

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approximately 4% on the DFC dataset and about 2.5% on the other datasets. The GSFE module improves the model's performance by reorganizing spatial information, enhancing rich spatial FE, and suppressing redundant spatial feature expression. The TokenLearner module also significantly contributes to performance improvement by enhancing semantic features and stabilizing central target features.

In the design of TokenLearner, we introduce the central target token $\boldsymbol{X}_{\mathrm{ct}}$. To verify its effectiveness, we test it under different patch sizes, with the experimental results shown in Table VII. At very small patch sizes, the introduction of $\boldsymbol{X}_{\mathrm{ct}}$ has minimal impact on the model and even slightly decreases performance on the IP, PU, and DFC datasets when the patch size is 5. This suggests that small patch sizes contain fewer pixels, reducing the likelihood of losing target information, hence the introduction of $\boldsymbol{X}_{\mathrm{ct}}$ may lead to redundant semantic information and becomes unnecessary. However, small patch sizes generally lack sufficient spatial information and are typically not utilized. As the patch size increases, the experimental results show that $\boldsymbol{X}_{\mathrm{ct}}$ plays a crucial role in enhancing model robustness. Larger patch sizes encompass more learnable information but also increase the risk of losing central target information, making the introduction of $\boldsymbol{X}_{\mathrm{ct}}$ necessary. Compared to other datasets, $\boldsymbol{X}_{\mathrm{ct}}$ has the least impact on the model's performance on the SA dataset, likely due to the extensive and continuous distribution of samples in this dataset, which makes the target information less prone to loss. The results demonstrate that the introduction of $\boldsymbol{X}_{\mathrm{ct}}$ is particularly beneficial in scenarios involving larger data augmentation sizes or where samples are scattered and discontinuous.

In the preprocessing stage, we apply the PCA algorithm to roughly rank the components while retaining all bands to avoid the loss of crucial discriminative information. To validate the effectiveness of this design, we conducted tests. The results in Table VIII show that this preprocessing method positively contributes to the stability of the model's performance. Due to limitations in data volume and computational resources, most HSI classification models are shallow networks. During the preprocessing stage, we perform a rough ranking of the band components to leverage prior knowledge and better guide shallow networks in distinguishing between primary and minor components, which facilitates subsequent FE. Further analysis reveals that for the first four datasets, which have a large number of bands but limited data, shallow networks face greater challenges in processing different spectral components. In this case, preprocessing has a more noticeable effect on stabilizing performance. However, on the DFC dataset, which contains only 48 bands but a relatively sufficient amount of data, the impact of preprocessing on model performance is relatively small. Given that HSI typically contains hundreds of spectral bands and available data is limited, we incorporate this preprocessing step in the model to enhance its stability.

V. Conclusion

In this article, we propose an HSI classification network called SSRRTL, which integrates the local FE capabilities of CNNs with the global modeling capabilities of transformer architectures. SSRRTL consists of the SRRMIE, GSFE, TokenLearner modules, and transformer encoders. The SRRMIE and GSFE modules are designed to effectively reduce spectral–spatial information redundancy in HSIs, preserving discriminative features and enhancing feature expression. The TokenLearner module facilitates transformer encoders in capturing semantic features of HSIs more effectively while retaining essential target information. In addition, SSRRTL incorporates multiscale information extraction and lightweight design considerations. Extensive experiments validate the outstanding performance of SSRRTL in various scenarios. In future work, we will attempt to introduce more diverse types and levels of semantic feature representation information into the TokenLearner module, and explore semi-supervised or unsupervised methods to reduce reliance on labeled samples.

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Jinbin Wu received the B.S. degree from Beijing Institute of Technology, Beijing, China, in 2021. He is currently pursuing the Ph.D. degree with the Department of Instrument Science and Engineering, School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China.

His research interests include hyperspectral image processing, deep learning, and signal processing.



Jiankang Zhao received the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2006.

He is currently a Professor with the Department of Instrument Science and Engineering, School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China. His research interests include image processing, bridge inspection, and satellite navigation.



Haihui Long received the bachelor's and master's degrees in applied mathematics from Central South University, Changsha, Hunan, China, in 2007 and 2010, respectively, and the Ph.D. degree in instrument science and technology from Shanghai Jiao Tong University, Shanghai, China, in 2015.

From 2015 to 2017, he was a Post-Doctoral Researcher with the Department of Automation, Shanghai Jiao Tong University, where he is currently a Lecturer with the Department of Instrument Science and Engineering. His research interests include robust control and intelligent control.

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